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Task Description:	Hematopoietic stem cells (HSC) comprise less than 0.1% of the bone marrow (BM) of adults, yet they produce all blood cells that are responsible for constant maintenance and immune protection of the body. In addition to the requirement for high cell output, the concentration of the different blood cell lineages produced are precisely controlled to allow not only preservation of normal homeostasis, but also the prompt response to specific stimuli and demands. Previous studies have suggested that exposure to galactic cosmic radiation (GCR) and solar particle event (SPE) radiation that occur or may potentially occur during space flight may increase the likelihood of carcinogenesis, with leukemias being one of the most frequent radiogenic cancers, and the ones with the smallest latency period. We hypothesize that GCR and SPE radiation can have a deleterious impact not only at the level of the mature immune cells and their committed progenitors, as is currently thought, but at the earlier developmental stage of the HSC. Genetic lesions induced by this exposure can lead directly to leukemic transformation, and may also cause these cells to generate a functionally compromised immune system lacking the ability to effectively fight challenges such as pre-leukemic transformations and viral/bacterial infections. As such, we hypothesize that astronauts would be doubly at risk, due to both the potential for induction of leukemia by SPE/GCR radiation, and the reduced ability of the astronauts' immune systems to identify and eliminate malignant cells or pathogens in the body. In the present proposal, we will use a variety of both in vitro cell culture and in vivo transplantation systems to rigorously test the novel hypothesis that combined SPE/GCR radiation can directly induce DNA damage and mutations within primitive HSC, and can also contribute to defective hematopoiesis/immunity through so-called tissue bystander effects, by likewise damaging the BM microenvironmental/niche cells (MSC) that regulate hematopoiesis. The overall goal of this proposal is, therefore, to utilize unique model systems developed in our laboratory to precisely and reproducibly define the effects of SPE/GCR radiation on the human hematopoietic and immune systems, at both the molecular and functional levels, thereby defining the possible risk of leukemogenesis as a result of exposure.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed studies directly address NASA's Radiation Program General Focus by studying cellular differentiation, genetic sensitivity or resistance, and immune function, all of which NASA has identified as being of particular importance. Moreover, the proposed approach of both creating a human BM microenvironment in immunodeficient mice and then repopulating that niche with human MSC and HSC that have been exposed to both SPE-like protons and low fluence heavy ion components of GCR directly addresses another of NASA's stated priorities which is "to develop new 3D human cell culture or transgenic mouse models of human tissues improving our ability to extrapolate estimates of cancer." With respect to the specific goals of this Announcement, our studies to employ human HSC and human BM microenvironmental cells directly address one of the stated high priority areas: "research that considers adult stem cells and the stem cell niche." In addition, our studies to investigate SPE/GCR-induced chromosomal aberrations (CAs), alterations in cancer-related SNPs and miRNAs, and mutations within coding sequences, assessed through whole exome sequencing (WES), directly respond to another stated high priority goal of elucidating "how genetic and epigenetic factors modulate biological responses to space radiation." Our intentional use of HSC and MSC samples from donors of both genders is directly responsive to NASA's statement "research on gender effects…are of interest." Thus, in combination, our proposed studies investigate multiple different cellular/molecular avenues of radiation carcinogenesis, and directly address another stated high priority for "research that will improve the understanding of radiation quality effects using two or more cancer development processes that are important in radiation carcinogenesis." With respect to NASA's identified "Radiation Carcinogenesis-Specific Gaps," our proposed studies address Cancer 1, as they include development of an experimental model of tumor development for leukemia, which, with this "humanized" model, will allow more direct extrapolation of human risk projection. The proposed genomics studies (CAs, miRNA, WES, and SNP arrays) and the inclusion of human donors of both genders (with an age range of typical astronauts) also directly addresses Cancer 5, which requests application of "models of cancer risk…to reduce the uncertainties in individual radiation sensitivity including genetic and epigenetic factors from GCR and SPE," and Cancer 6, which requests application of "models of cancer risk… to reduce the uncertainties in age and gender dependence of cancer risks from SPEs and GCR."
	During the past year of this project, we successfully brought human marrow stromal cells (MSC) and hematopoietic stem and progenitor cells (HSC) from additional human donors to that NASA Space Radiation Laboratory (NSRL) 3 times, irradiated these cells with either cesium-137 gamma radiation (1 Gy, 662 keV) at Brookhaven National Laboratory (BNL) Biosciences or SPE/GCR radiation at NSRL, and brought these cells back to Wake Forest Institute for Regenerative Medicine (WFIRM) without contamination, and with no negative impact on viability or functionality of either cell type. We again performed single ion proton (1 Gy, 50 MeV; simulated SPE) exposures, single ion 56Fe (20 cGy, 1 GeV; simulated GCR) exposures, and sequential proton + iron exposures, and also began investigating the efficacy of a readily available dietary supplement to serve as a countermeasure to the deleterious effects of gamma, proton, and 56Fe ion radiation on the human hematopoietic system. The majority of this year's work was focused on three key aspects of the proposed studies: 1) testing of a dietary supplement to protect against/mitigate the damaging effects of sSPE and/or GCR ions on human HSC function; 2) genomic/transcriptomic analysis of the HSC and MSC following exposure to the aforementioned radiation schemes; and 3) in vivo studies in which human HSC that had been exposed to simulated SPE and/or GCR radiation were transplanted into immunodeficient mice to repopulate the hematopoietic system with these potentially damaged HSC clones, and define alterations in engraftment and differentiation, and assess the risk of leukemogenesis. 1) Testing of a novel Countermeasure: To assess the ability of a common, inexpensive dietary supplement (which could easily be administered to astronauts during extended missions) to protect against and/or mitigate the deleterious effects of sSPE and sGCR on human HSC functions, we isolated human HSC as in prior years, and preincubated them with this agent prior to irradiation (employing each of the above detailed schemes), or treated with the agent within 1.5 hours after irradiation. HSC were then returned to WFIRM, and the functionality assayed in vitro using the HALO®-96 Human Stem/Progenitor Cell Assay (HuSPCA, Hemogenix), based on our success with this system during NSRL 14A, B, and C. The brief preincubation of human HSC with this agent prior to irradiation completely protected the most primitive HPP-CFC from the pronounced deleterious effects of gamma and 56Fe ions (actually increasing the HPP-CFC potential beyond that seen in the unirradiated controls), but had little impact on the negative effects protons exerted on HPP-CFC potential. Importantly, this agent was also able to protect the functionality of the most primitive HPP-CFC

Task Progress:	when administered after exposure to gamma rays, 56Fe ions, and to some extent, protons and sequential protons followed by 56Fe ions. Looking at the effects gamma, sSPE, and GCR ions had on the ability of human HSC to give rise to the somewhat more differentiated CFU-GM colonies, this agent was highly protective if administered prior to exposure to gamma, protons, or 56Fe ions, and was able to partially restore CFU-GM forming potential (to levels of about 60% of unirradiated controls) even when given after radiation exposure. Based on these highly promising results, studies are being planned to assess the ability of this dietary supplement to restore engraftment and differentiation ability of sSPE and/or GCR ion-exposed human HSC when transplanted into immunodeficient mice. 2) SPE/GCR-Induced Alterations to in vivo Functionality of Human HSC: While in vitro assays can provide valuable information regarding the functionality of human HSC, the gold standard for assessing the true potential/function of human HSC is to test their ability to engraft and repopulate the hematopoietic system following transplantation into conditioned/ablated immunodeficient (to allow engraftment of the human cells) murine recipients. We therefore transplanted immunodeficient (NSG) mice with human HSC that had been exposed to one of the following irradiation schemes (n=9 per group) at the NSRL: 1) sham irradiation; 2) 1 Gy of 137Cs gamma-irradiation; 3) 1 Gy of 50 MeV protons (SPE); 4) 20 cGy of 1 GeV/n 56Fe ions (GCR); or 5) 1Gy of 50 MeV protons followed immediately by 20 cGy of 1 GeV/n 56Fe ions. Mice were then sacrificed at 4-6 months post-transplant, their spleen and bone marrow harvested, and analyzed by multicolor flow cytometry to ascertain the levels of human hematopoietic engraftment and assess whether exposure to SPE/GCR altered the engraftment potential and/or differentiative capacity of the HSC. Our results demonstrate that treatment of human HSC with gamma-rays, protons, 56Fe ions, or protons followed immediately by 56Fe ions ("mix") markedly altered the ability of the human HSC to mediate long-term (5-6 months) engraftment, and led to pronounced skewing of the differentiative potential of the human HSC. Of particular note is the significant impairment we observed in commitment to the erythroid lineage following exposure of human HSC to all radiation types tested. When we examined the spleen of the mice that were repopulated with human HSC exposed to gamma-rays, protons, 56Fe ions, or "mix", a reduction in erythropoiesis mirrored that which we had seen in the bone marrow. However, there was also a marked expansion of the myeloid lineage as a result of exposing the HSC to each radiation type prior to transplant. In addition, two of the recipients of 56Fe ion-irradiated human HSC exhibited dramatic splenomegaly, with one mouse's spleen reaching ~30X the normal size, which is highly suggestive of the development of a hematological malignancy. Cytogenetic analysis is currently being performed on cells from this mouse that were cryopreserved at the time of sacrifice. In addition, high-resolution microscopic images of 10,000 human cells from this mouse's spleen have been acquired/captured using an Amnis ImageStream imaging flow cytometer, and are currently being analyzed using Amnis' IDEAS software package, and suggest that large, blast-like cells are present in the spleen of this mouse. Secondary transplants of human cells from this mouse into a group of new NSG recipients to better assess leukemogenic potential are soon to be underway. 3) Transcriptomic/Proteomic/Genomic Analyses: As discussed in last year's progress report, we collected culture supernatants from the MSC from each donor at 48 hours following exposure to each type of radiation, using the culture supernatant from the MSC that were flown to NSRL and not exposed to radiation as donor-specific controls. We then analyzed these supernatants using a luminescence-based Human Cytokine Array, which allows simultaneous quantitation of the expression levels of 42 cytokines involved in hematopoiesis/inflammation/immune function to begin delineating the mechanism whereby the exposure of MSC to each form of radiation might alter the ability of the MSC to support normal/functional hematopoiesis. These assays have thus far revealed marked alterations in multiple pathways of direct relevance to MSC's ability to support homing/retention of HSC within the bone marrow niche, HSC self-renewal, and HSC differentiation into functional immune cells. In addition to performing the aforementioned cytokine arrays, we also collected cell pellets from the MSC and from the hematopoietic cells from each treatment group. We isolated RNA from the MSC and DNA from the HSC for analysis. The mRNAs from the MSC were analyzed by Illumina microarrays, and the DNA from the purified human HSC was subjected to whole exome sequencing (WES). We have now begun processing all of this data to compile a comprehensive picture of what changes each form of radiation, singly or in combination, produces at both the level of the genome (i.e., mutations, deletions, translocations) and the transcriptional level. Thus far, our results have demonstrated very consistent effects from one donor to another, and have shown that each type of radiation produces a very distinct effect on the transcriptional program of the human MSC, with the gene expression pattern clustering very tightly based upon radiation type. We also found that the epigenetic changes induced by protons and 56Fe ions were far more stable than those induced by traditional gamma radiation, since the majority of the changes in response to gamma radiation were lost as the MSC were propagated in culture, while those induced by exposure to protons or 56Fe ions persisted within the daughter cells following multiple cell divisions. The exome sequencing of the human HSC from 2 donors have provided a wealth of information on the genomic alterations that occur within these cells as a result of exposure to gamma rays, protons, and 56Fe ions. In similarity to the results seen with RNA expression in the MSC, initial analysis of the data from these two donors indicates that the response in both donors is remarkably similar. Deep, in-depth mining of these data is now ongoing to catalog the various genomic alterations/damage induced by each radiation type within the HSC. This will be the first time information of this nature has been available in the human system, making it possible to begin assessing the precise risks GCR and SPE radiation may pose to the astronauts' hematopoietic system.
Bibliography Type:	Description: (Last Updated: 08/05/2025)
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